Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**LAB – Reinforcement Learning**
Semester	6th	**Course Code**	**23ADS1644**
Roll No	**42**	**Name of Student**	**Krishna Mishra**

Practical Number	**05**
Aim	To implement a multi-armed bandit problem to study the exploration-exploitation trade-off.
Problem Definition	To implement a multi-armed bandit model to analyze and balance the exploration-exploitation trade-off for maximizing cumulative reward.
Theory (100 words)	The Multi-Armed Bandit problem is a fundamental concept in reinforcement learning that models decision-making under uncertainty. It is based on a scenario where a player chooses between multiple options (arms), each providing a random reward. The main objective is to maximize total reward over time by balancing exploration and exploitation. Exploration involves trying different actions to gather information about their rewards, while exploitation focuses on selecting the action known to yield the highest reward. Achieving an optimal balance between these two strategies is crucial for efficient learning. Common approaches to solving this problem include the epsilon-greedy method, where a small probability is used to explore random actions, and the remaining probability is used to exploit the best-known option. Other advanced methods include Upper Confidence Bound (UCB) and Thompson Sampling. The multi-armed bandit framework is widely used in real-world applications such as online advertising, recommendation systems, and clinical trials, where decisions must be made sequentially with limited information.

Procedure and Execution

(100 Words)

Algorithm:

1. Import required libraries (NumPy, Matplotlib).
2. Define the number of bandit arms and their reward distributions.
3. Initialize variables to store rewards, counts, and estimated values.
4. Choose a strategy (e.g., epsilon-greedy).
5. Set exploration probability (epsilon).
6. For each iteration, decide whether to explore or exploit.
7. Select an arm, receive reward, and update estimates.
8. Track cumulative reward and visualize performance.
9. Analyze results.

Code:

```
# Import libraries
import numpy as np

class EpsilonGreedy:
    def __init__(self, n_arms, epsilon):
        self.n_arms = n_arms
        self.epsilon = epsilon
        self.counts = np.zeros(n_arms)
        self.values = np.zeros(n_arms)

    def select_arm(self):
        if np.random.rand() < self.epsilon:
            return np.random.randint(0,
self.n_arms)
        else:
            return np.argmax(self.values)

    def update(self, chosen_arm, reward):
        self.counts[chosen_arm] += 1
```

```
        n = self.counts[chosen_arm]
        value = self.values[chosen_arm]
        self.values[chosen_arm] = ((n - 1) /
n) * value + (1 / n) * reward

# Parameters
n_arms = 10
epsilon = 0.1
n_trials = 1000

# True rewards
true_means = np.random.randn(n_arms)
print(f"Actual mean rewards:
{np.round(true_means, 2)}")
print(f"Best arm: {np.argmax(true_means)} with
mean {np.max(true_means):.2f}\n")

# Initialize agent
agent = EpsilonGreedy(n_arms, epsilon)
total_reward = 0

# Run trials
for t in range(n_trials):
    arm_to_pull = agent.select_arm()
    reward = np.random.randn() +
true_means[arm_to_pull]
    agent.update(arm_to_pull, reward)
    total_reward += reward

print(f"Total Reward: {total_reward:.2f}")
```

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	Output: Actual mean rewards: [-2.3 0.77 -2.25 -0.1 -1.43 0.58 2.49 -0.38 -0. -0.62] Best arm: 6 with mean 2.49 Total Reward: 1990.56
Conclusion	The multi-armed bandit approach effectively demonstrates the exploration-exploitation trade-off. It helps in making optimal decisions over time by balancing learning and reward maximization in uncertain environments.
Date	**28-03-2026**

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